MUHAMMAD IDHAM BIN ISA

AI & IT Engineer

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GitHub: github.com/idhamisa   LinkedIn: linkedin.com/in/idham-isa

Portfolio: idhamisa.github.io

TECHNICAL SKILL

- Languages: Python, C++, JavaScript
- Frameworks: Django, Flask, FastAPI, Node.js, Express
- AI/CV: Deep Learning, Computer Vision, **CNN**, **YOLO**, OpenCV
- Optimization: TensorFlow Lite, ONNX, OpenVINO
- Databases: MongoDB, SQLite
- Embedded: Raspberry Pi, STM32, Arduino, ESP32

WORK EXPERIENCE

AI & IT Engineer, PK Agro Industrial Products (M) Sdn. Bhd — **Dec. 2025 - Present**

- Developed **AI-based computer vision systems** to automate manual inspection processes in factory operations
- Built **real-time detection pipelines** using **YOLO, OpenCV, and Python**
- Integrated **IP camera (RTSP)** for live monitoring (Dahua and Hikvision camera)
- Developed backend systems using Django / Flask
- Optimized models using **Pytorch** and **OpenVINO** for deployment
- Improved **operational efficiency** through automated detection and monitoring

Internship, Melaka ICT Holding Sdn. Bhd. — **Jun. 2024 - Sep. 2024**

- Assisted in website content updates and basic maintenance using **Wordpress**.
- Supported network setup by connecting and organizing Ethernet cables for server operations.
- Designed digital and print materials to support marketing and communication needs.

PROJECTS

AI-Based Bin Monitoring System

- Developed **YOLO-based object detection** system for bin level monitoring
- Built backend APIs using **Flask & Django** for real-time data processing
- Integrated **CCTV** systems (**Dahua/Hikvision** RTSP streams)
- Optimized inference performance for real-time processing
- Addressed challenges in lighting variation and camera angles using preprocessing techniques

Final Year Project – PLC Rejection Integrated System Based on Deep Learning

- Designed and implemented an automated rejection system for can cap defects using PLC and AI integration.
- Applied **CNN** and **YOLO** for real-time defect detection and classification.
- Deployed AI models on **Raspberry Pi 5** with Hailo AI Hat for high-speed processing.
- Controlled **PLC actuators** to reject defective can caps based on AI output.

Roadtax Expiry Detector

- Developed an AI-powered Road tax Expiry Detector using **OCR** and computer vision.
- Built a system to capture vehicle number plates, extract text, and identify road tax expiry dates.
- Implemented automated alerts to notify users when expiry dates were approaching.
- Designed the project to enhance vehicle management automation and support compliance monitoring.

EDUCATION

Bachelor of Computer Engineering with Honours **Sep. 2021 - Jul. 2025**

Universiti Teknikal Malaysia Melaka(UTeM)

- CGPA: 3.35, Dean's List: 2 Semesters
- Relevant Course: Computer Vision and Pattern Recognition, Machine Learning, Artificial Intelligent

Matriculation (Biology, Physics, Chemistry) **Apr. 2020 - May 2021**

Johor Matriculation College

- CGPA: 3.34, Dean's List: 1 Semester

CO-CURRICULAR ACTIVITIES & ACHIEVEMENTS

- Gold Medalist – INOTEK 2025 (Final Year Project Presentation)
- Bronze Medalist – INOTEK 2024 Festival (Integrated Design Project)
- Participant – Vision Showcase 2025 (Artificial Intelligence Project Showcase)
- Volunteer – SULAM Program: Taught microcontroller & coding to secondary school students
- Crew Member – Haunted House Event, FTKEK Festival (2022 & 2023)

CERTIFICATION AND TRAINING

- **Omron PLC level 1 and level 2 Certification**

LANGUAGES

- **Bahasa Malaysia:** Fluent
- **English:** Intermediate

REFERENCES

References available upon request